UNIFORMLY EMBEDDABLE POLARIZED OBJECTS FORM A NOETHERIAN SUBSET

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2128-2144 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 2146-2204, the complete author proof (57 body lines).

Editorial note. Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 5 of queue_moduli.json (pb=2359, stmt 2128-2144, proof 2146-2204).

Editorial note. stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Editorial note. Author footnote '(insert future reference here)' is a todo annotation on a citation-improvement remark; it does not touch the argument chain.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DPH: sources/stacks-moduli/pinned-original-source.tex, source lines 1456–1466. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DPU: sources/stacks-moduli/pinned-original-source.tex, source lines 2112–2115. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0D3W: sources/stacks-moduli/pinned-original-source.tex, source lines 2204–2204. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E94: sources/stacks-moduli/pinned-original-source.tex, source lines 2204–2204. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:moduli.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0DQI: sources/stacks-moduli/pinned-original-source.tex, source lines 2204–2204. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06R6: sources/stacks-moduli/pinned-original-source.tex, source lines 2204–2204. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0052: sources/stacks-moduli/pinned-original-source.tex, source lines 2204–2204. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0053: sources/stacks-moduli/pinned-original-source.tex, source lines 2204–2204. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $n \geq 1$ be an integer and let P be a numerical polynomial. Let*

$$T \subset |\mathit{Polarized}|$$

be a subset with the following property: for every $\xi \in T$ there exists a field k and an object $(X, \mathcal{L})$ of $\mathit{Polarized}$ over k representing ξ such that

(1) *the Hilbert polynomial of $\mathcal{L}$ on X is P, and*
(2) *there exists a closed immersion $i : X \to \mathbf{P}^n_k$ such that $i^*\mathcal{O}_{\mathbf{P}^n}(1) \cong \mathcal{L}$.*

Then T is a Noetherian topological space, in particular quasi-compact.

Proof. Observe that $|\mathit{Polarized}|$ is a locally Noetherian topological space, see Morphisms of Stacks, Lemma 0DQI (this also uses that $\mathrm{Spec}(\mathbf{Z})$ is Noetherian and hence $\mathit{Polarized}$ is a locally Noetherian algebraic stack by Lemma 0DPU and Morphisms of Stacks, Lemma 06R6). Thus any quasi-compact subset of $|\mathit{Polarized}|$ is a Noetherian topological space and any subset of such is also Noetherian, see Topology, Lemmas 0053 and 0052. Thus all we have to do is a find a quasi-compact subset containing T.

By Lemma 0DPH the algebraic space

$$H = \mathrm{Hilb}^{P,\mathcal{O}(1)}_{\mathbf{P}^n_{\mathbf{Z}}/\mathrm{Spec}(\mathbf{Z})}$$

is proper over $\mathrm{Spec}(\mathbf{Z})$. By Quot, Lemma 0D3W[1] the identity morphism of H corresponds to a closed subspace

$$Z \subset \mathbf{P}^n_H$$

which is proper, flat, and of finite presentation over H and such that the restriction $\mathcal{N} = \mathcal{O}(1)|_Z$ is relatively ample on Z/H and has Hilbert polynomial P on the fibres of $Z \to H$. In particular, the pair $(Z \to H, \mathcal{N})$ defines a morphism

$$H \longrightarrow \mathit{Polarized}$$

which sends a morphism of schemes $U \to H$ to the classifying morphism of the family $(Z_U \to U, \mathcal{N}_U)$, see Quot, Lemma 0E94. Since H is a Noetherian algebraic space (as it is proper over $\mathbf{Z}$)) we see that $|H|$ is Noetherian and hence quasi-compact. The map

$$|H| \longrightarrow |\mathit{Polarized}|$$

[1]We will see later (insert future reference here) that H is a scheme and hence the use of this lemma and Quot, Lemma 0E94 isn't necessary.

is continuous, hence the image is quasi-compact. Thus it suffices to prove T is contained in the image of $|H| \to |\mathcal{P}olarized|$. However, assumptions (1) and (2) exactly express the fact that this is the case: any choice of a closed immersion $i : X \to \mathbf{P}^n_k$ with $i^*\mathcal{O}_{\mathbf{P}^n}(1) \cong \mathcal{L}$ we get a k-valued point of H by the moduli interpretation of H. This finishes the proof of the lemma. □